

LESSON 4: AI IN THE REAL-WORLD

Artificial Intelligence (AI) is no longer just a futuristic idea from science fiction. It is already part of our everyday lives, often in ways we do not even notice. From recommending what video to watch next to helping doctors detect diseases earlier, AI systems are shaping how people live, work, and learn.

In this lesson, we will explore how AI is used in the real world, what challenges and limitations it faces, and why ethics and responsibility are central to its development. By the end, you should be able to explain examples of AI applications, recognize where AI might fail, and discuss the impact AI has on individuals and society.

APPLICATIONS OF AI

AI technologies are designed to extend the very processes of intelligence we studied in earlier lessons. They can recognize patterns in data, form associations between pieces of information, and learn from past examples. With these abilities, AI systems can make predictions, classify new situations, and even carry out forms of reasoning such as inference, decision-making, and explanation. Because these are the same foundations of human intelligence, AI can be applied across many areas of daily life, industry, and science.

Most of today's systems fall under Narrow AI, which means they are designed with a narrow focus. They can perform a single task or a limited set of tasks very effectively. Examples include navigation apps that suggest the fastest route or chatbots that respond to customer service questions. Although they may appear intelligent, their abilities are restricted to their specific domain and cannot be transferred to entirely different tasks.

NARROW AI SYSTEMS ARE DESIGNED TO PERFORM ONE SPECIFIC TASK OR A LIMITED SET OF TASKS VERY EFFECTIVELY.

By contrast, researchers often speak of General AI (Artificial General Intelligence, AGI). General AI is still theoretical—it has not yet been created. In principle, a General AI would have the ability to learn and apply intelligence across many tasks, just as humans can. It could transfer knowledge from one situation to another, adapt to unfamiliar challenges, and reason flexibly in new contexts. In other words, it would be capable of applying its intelligence to virtually any domain.

GENERAL AI COULD LEARN AND APPLY INTELLIGENCE FLEXIBLY
ACROSS MANY DIFFERENT TASKS.

Narrow AI applications, on the other hand, are typically classified according to the kind of cognitive function they mimic. The five most widely recognized categories are:

1. Perception
2. Language
3. Reasoning and Decision-Making
4. Prediction and Forecasting
5. Control and Robotics

PERCEPTION

Perception is the stage where AI systems handle raw input from the world. Just as humans use their senses to see and hear, AI uses sensors, cameras, and microphones to detect signals. The goal is to transform pixels, sound waves, or other raw data into information the system can work with.

PERCEPTION IS THE ABILITY OF AI TO INTERPRET SENSORY INPUT,
ESPECIALLY VISION AND SOUND.

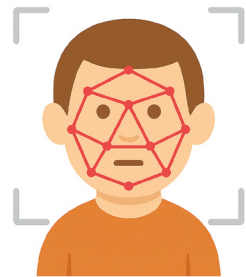
Computer vision

Computer vision allows AI to recognize and analyze visual information. This includes identifying faces in photos, detecting objects in security footage, and even reading handwriting.

For example, in medicine, AI systems are trained to analyze X-rays, CT scans, and MRI images, highlighting areas that might contain signs of disease. Farmers use drones with computer vision to monitor crop health, spotting areas affected by pests or lack of water. Retailers rely on AI-powered cameras to track how shoppers move through stores, helping them redesign layouts to improve shopping experiences. Many smartphones now use facial recognition as a security feature.

Examples of real-world applications:

- Identifying faces in photos
- Detecting objects in security footage
- Reading handwriting
- Analyzing X-rays, CT scans, and MRI images in medicine
- Monitoring crop health with drones in agriculture
- Tracking shopper movement in retail stores
- Smartphone facial recognition for security



Speech recognition

Speech recognition is a branch of AI perception. It allows AI to convert spoken language into digital signals that can later be processed as text. This technology makes it possible for computers to “hear” human speech. Speech recognition itself is not responsible for understanding or acting on the speech, but it provides the raw input that other AI systems require.

For example, systems like Siri, Alexa, and Google Assistant rely on speech recognition to convert spoken words into text before carrying out commands.

Automatic captioning tools also depend on speech recognition to generate subtitles. In some applications, speech recognition works together with language processing, such as in real-time translation between languages.

Examples of real-world applications:

- Virtual assistants (Siri, Alexa, Google Assistant)
- Automatic captioning tools for videos
- Real-time translation apps

**Limitations & Challenges**

Although AI perception has made impressive progress, it does not “see” or “hear” in the way humans do. Instead, it processes raw data such as pixels in an image or sound waves in an audio file. This allows AI systems to recognize patterns quickly and sometimes with remarkable accuracy, but there are important limitations:

- **Task-specific training:** An AI trained to recognize cats in photos cannot automatically recognize dogs without specific training, just as a speech recognition system trained on one language cannot understand another language without retraining. Each new skill requires its own dataset and training process.
- **Lack of common sense:** AI may detect shapes or sounds but does not understand their meaning in context—for example, it might confuse a toy cat with a real one.
- **Data dependence:** Performance depends heavily on the quality and diversity of the data. If the training data is incomplete or biased, the system may make mistakes in real-world situations.
- **Fragility:** Small changes in input, such as unusual lighting in an image or background noise in audio, can cause AI systems to fail where humans would still succeed.

LANGUAGE

Language is one of the most complex human abilities, involving not only words but also meaning, context, and culture. AI has made remarkable progress in this field through Natural Language Processing (NLP), which allows computers to process and generate human language in both written and spoken forms.

NATURAL LANGUAGE PROCESSING ANALYZES, TRANSLATES, AND GENERATES HUMAN LANGUAGE IN TEXT OR SPEECH.

Real-world applications of AI in language include:

- Translation tools (e.g., Google Translate) that enable instant communication across languages.
- Chatbots that provide customer service by analyzing questions and generating responses.
- Large language models (e.g., ChatGPT, Gemini) that can summarize documents, draft essays, or even write computer code.
- Content filtering on social media platforms to detect spam, offensive, or inappropriate material.
- Sentiment analysis that helps businesses understand customer opinions by scanning reviews and posts.
- Educational tools that give feedback on student writing, correcting grammar and suggesting improvements.



In all these applications, Natural Language Processing (NLP) specifically handles the processing of language—analyzing, interpreting, and generating text. These systems often work alongside speech recognition, which first converts spoken words into text before NLP can take over.

Examples of real-world applications:

- Translation tools
- Chatbots
- Large language models
- Content filtering on social media platforms
- Sentiment analysis
- Educational tools

**Limitations & Challenges**

AI language systems can translate text, summarize information, or generate conversations, but they face several challenges:

- Shallow understanding: AI does not truly “understand” meaning; it relies on statistical patterns in data, which can lead to irrelevant or confusing answers.
- Bias in training data: Language models can reproduce stereotypes or offensive language if such patterns are present in the data.
- Difficulty with context: Long conversations, sarcasm, humor, and cultural references are often misunderstood.
- Errors in translation: While real-time translation is powerful, idioms and cultural expressions are frequently mistranslated.

These issues show that AI language tools are valuable helpers but cannot replace human judgment or cultural awareness. They are best used as support systems that speed up tasks, provide suggestions, or make communication easier, while leaving interpretation and final decisions to people. Understanding their limitations helps users apply them wisely and avoid overreliance. In this way, AI language tools can enhance human abilities rather than attempt to replace them.

REASONING AND DECISION-MAKING

Reasoning involves drawing conclusions from information and making choices based on those conclusions. In AI, reasoning and decision-making represent the choice stage, where the system evaluates options and selects an action. While humans rely on experience and common sense, AI applies rules, models, or learned patterns to make choices within a specific domain. Although limited in scope, AI reasoning can be highly effective when focused on a well-defined task.

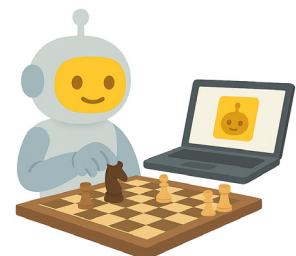
REASONING AND DECISION-MAKING ARE AI'S ABILITY TO EVALUATE OPTIONS AND CHOOSE ACTIONS WITHIN A SPECIFIC DOMAIN.

AI reasoning appears in many areas of daily life. Recommender systems are one of the most common examples. Platforms like YouTube, Netflix, or Spotify analyze a user's past behavior and suggest what to watch or listen to next. Online retailers such as Amazon use similar methods to recommend products based on browsing and purchase history.

In logistics and planning, delivery companies apply AI to calculate the most efficient routes for trucks, balancing distance, fuel, and time. Airlines rely on AI systems to adjust ticket prices depending on demand. Another striking example is in game-playing, where AIs that defeated world champions in chess and Go used reasoning strategies to explore possible moves and select the most promising path.

Examples of real-world applications:

- Recommender systems to suggest videos and music
- Online retailers to recommend products
- Logistics systems to calculate the most efficient route
- Airlines to adjust ticket prices
- Game-playing AIs to choose the best moves



In conclusion, AI reasoning is powerful when applied to narrow, clearly defined problems. It can increase efficiency, improve recommendations, and solve complex challenges in ways that humans alone may find difficult.

Limitations & Challenges

Despite its strengths, AI reasoning faces several important limitations:

- **Domain restriction:** AI can only reason within the data and rules it has been trained on. It cannot transfer decision-making skills to unrelated tasks.
- **Lack of common sense:** Algorithms may choose mathematically optimal solutions that are impractical or unsafe in real life.
- **Hidden bias:** Decisions made by AI in areas such as hiring or lending may reflect biases in the training data, leading to unfair outcomes.
- **Transparency:** Many decision-making systems are “black boxes,” making it difficult to explain how or why a conclusion was reached.

These challenges raise concerns about trust and accountability, especially in areas like finance, healthcare, and law enforcement. In these fields, even small mistakes or hidden biases can cause serious harm. If people cannot see how AI makes its decisions, it is hard to question or fix unfair outcomes. This makes it important to keep AI systems fair, clear, and supervised by humans.

Notes

Common Misconceptions

- *AI can think like humans.*

AI does not understand or think — it recognizes patterns and follows instructions.

- *AI is always right.*

AI can make mistakes, especially if the data is biased, incomplete, or poor quality.

PREDICTION AND FORECASTING

Prediction is one of AI's greatest strengths. By analyzing large amounts of past data, AI can identify patterns and extend them into the future.

PREDICTION AND FORECASTING USE PAST AND CURRENT DATA TO IDENTIFY PATTERNS AND PROJECT FUTURE OUTCOMES.

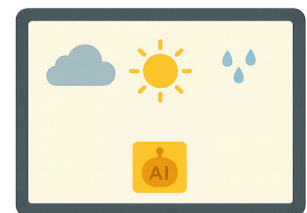
Some applications are now highly established. In finance, AI forecasts stock trends and detects suspicious transactions that may indicate fraud. In healthcare, predictive systems estimate the likelihood of disease outbreaks and assess patient risks. In meteorology, AI improves weather forecasts by analyzing satellite images and atmospheric data. And in retail and e-commerce, predictive tools anticipate customer demand, manage inventory, and reduce waste.

Other applications are still emerging. In sports, teams use AI to analyze player performance and suggest training strategies. In education, predictive analytics helps teachers identify students who may be at risk of falling behind. Governments are also beginning to apply predictive AI in urban planning, such as analyzing traffic flows or planning energy use.

These applications are powerful, yet they also have limits. Predictions are never perfect; their accuracy depends on the quality of the data. For example,

Examples of real-world applications:

- Finance to forecast stock trends
- Healthcare to predict disease outbreak
- Meteorology to generate weather forecasts
- Retail and e-commerce to anticipate demand
- Sports to analyze player performance
- Education to identify students at risk of falling behind
- Governments to predict traffic and energy needs



an AI trained only on sunny weather cannot perform well in a rainy climate. And without human oversight, relying too heavily on predictions can lead to costly mistakes.

Limitations & Challenges

AI prediction relies on patterns in past data, which creates important challenges:

- **Dependence on data quality:** Predictions are only as good as the data used to train the system. If the data is incomplete, biased, or inaccurate, the predictions will also be flawed.
- **Inability to handle the unexpected:** AI struggles with rare or new events that never appeared in its training data, so it may fail in unusual situations.
- **Overconfidence:** Predictions often look exact, but even small errors can grow into big problems, such as in financial trading or weather forecasting.
- **Ethical risks:** Predictive systems in areas like policing or education may unfairly label people, which can limit their opportunities and create long-term harm.

Predictions are powerful tools that can guide decisions, but they should always be checked and supervised by humans.

Notes

Common Misconceptions

- *AI will replace humans completely..*

AI can automate tasks, but it still requires human oversight, creativity, and judgment.

- *Robots and AI are the same thing.*

AI is software (algorithms and models), while robots are machines that may use AI to act.

CONTROL AND ROBOTICS

Control refers to the ability of AI to direct machines that act in the physical world. Robotics combines perception, reasoning, and prediction into coordinated action.

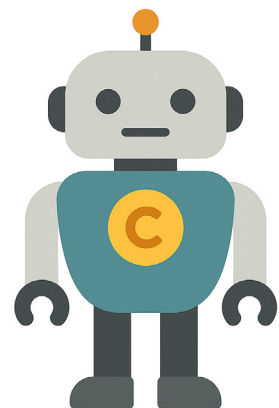
CONTROL AND ROBOTICS IS THE USE OF AI TO DIRECT MACHINES THAT ACT IN THE PHYSICAL WORLD.

One of the most familiar examples of AI in control is the self-driving car. These vehicles use cameras and sensors to take in information about their surroundings. AI perception helps them “see” the road, traffic signs, and obstacles. Prediction is used to guess how other drivers or pedestrians might move, and reasoning helps the car decide when to change speed, stop, or turn.

Robots in industry are another important use of AI. In factories, robotic arms assemble products with speed and precision, often working faster and more accurately than humans on repetitive tasks. Delivery robots and drones also rely on AI to plan safe routes and avoid obstacles, whether they are navigating sidewalks, warehouses, or the air.

Examples of real-world applications:

- Self-driving cars to navigate roads safely
- Industrial robots to assemble products in factories
- Delivery robots to carry goods through warehouses
- Drones to deliver packages
- Robotic surgery to assist doctors with operations
- Agricultural robots to harvest crops and sort produce
- Disaster-response robots to search for survivors in dangerous areas



AI-controlled robots are also becoming more common in other fields. In healthcare, robotic surgery uses AI to guide surgical instruments with great precision, which can shorten recovery times for patients. In agriculture, robots can harvest crops, sort produce, or spray fields, helping farmers reduce waste and save time. In disaster response, AI-driven robots are being tested to search for survivors in collapsed buildings, flooded areas, or other dangerous environments where sending humans would be too risky.

In all these cases, AI is not just analyzing information but also acting in the physical world, turning data into movement and decisions. This shows how AI control and robotics bring together perception, prediction, and reasoning to create systems that can work alongside people in many different settings.

Limitations & Challenges

AI systems that act in the physical world combine perception, reasoning, and prediction, but they are far from perfect:

- **Limited adaptability:** Robots trained for one task, such as picking strawberries, cannot easily adapt to another, such as harvesting apples.
- **Environmental challenges:** Self-driving cars may work well in one city but struggle in different weather, traffic, or road conditions.
- **Mechanical and safety risks:** Hardware failures or incorrect decisions can lead to dangerous accidents.
- **High costs:** Building and maintaining advanced robots is expensive, which limits widespread adoption.

These challenges remind us that while robots and AI-controlled machines can perform repetitive or risky tasks, they still require human supervision and intervention.

ETHICS AND RESPONSIBILITY

As AI becomes more common in daily life, questions of ethics—what is right or wrong—become increasingly important. Ethical considerations guide how AI should be designed, used, and supervised so that it benefits society while avoiding harm.

WHY ETHICS MATTERS

AI is powerful because it extends core human abilities—perception, language, reasoning, prediction, and control—into machines. But power comes with responsibility. Without ethical guidelines, AI could be misused, cause harm, or increase inequality.

For example, if a healthcare AI is trained only on data from adults, it may give poor advice when used with children, creating unfair outcomes. A predictive policing system that relies on biased crime data may target certain communities more than others, reinforcing social injustice.

Ethics ensures that AI systems are designed and used in ways that respect human rights and dignity. It reminds us to ask questions such as: *Who benefits from this technology? Who might be harmed? How can risks be reduced?*

Ethical considerations also help build trust. If people believe that AI is fair, transparent, and accountable, they are more likely to accept and use it. If they do not, society may reject valuable technologies out of fear or suspicion.

In short, ethics matters because AI does not exist in isolation—it affects real people, influences real decisions, and shapes the future of communities worldwide. By applying ethical principles, we can guide AI to support fairness, safety, and positive social change.



UNETHICAL CASE STUDIES

- ***Bias and Fairness***



Case Study: A university used an AI system to help rank student applications. Because the system was trained mostly on past data from high-income schools, it gave unfair preference to wealthy students while disadvantaging equally qualified students from less privileged backgrounds. This shows how bias in data can lead to unfair results.

- ***Privacy***



Case Study: A major social media company secretly used AI to scan private messages in order to target users with ads. Many people felt their personal conversations had been monitored without consent, which caused public backlash and loss of trust. This highlights why privacy is a key ethical concern in AI.

- ***Transparency and Accountability***



Case Study: In the U.S., some courts used an AI tool called COMPAS to predict how likely defendants were to reoffend. Judges used these scores in bail, parole, and sentencing decisions. Researchers later found the system gave harsher scores to Black defendants than to white defendants, and because its reasoning was hidden, unfair outcomes were hard to challenge.

- ***Dependence and Displacement***



Case Study: A factory introduced AI-controlled robots to increase production speed. While profits rose, hundreds of workers lost their jobs without retraining opportunities. This created unemployment and hardship in the local community, showing how AI can displace workers if changes are not managed responsibly.

SUMMARY

Narrow AI vs. General AI

- **Narrow AI:** Current real-world AI; designed for specific tasks (e.g., translation, face recognition). Cannot transfer skills to new tasks.
- **General AI:** A future goal; would learn and apply intelligence across many tasks like a human. Not yet achieved.

Five Cognitive Categories of AI

- **Perception (Input stage):** Converts raw data (images, sounds) into usable information. Examples: computer vision, speech recognition.
- **Language (Processing stage):** Interprets and generates human language. Examples: translation tools, chatbots, content filtering.
- **Reasoning & Decision-Making (Choice stage):** Evaluates options and selects actions. Examples: recommender systems, logistics planning, dynamic pricing.
- **Prediction & Forecasting (Future stage):** Uses past data to estimate future outcomes. Examples: stock forecasts, weather models, education risk alerts.
- **Control & Robotics (Action stage):** Directs machines to act in the physical world. Examples: self-driving cars, industrial robots, robotic surgery.

Limitations & Challenges

- **Perception** – Task-specific training, lack of common sense, data dependence, fragility
- **Language** – Shallow understanding, bias in data, context difficulty, translation errors
- **Reasoning and Decision-Making** – Domain restriction, lack of common sense, hidden bias, transparency issues
- **Prediction and Forecasting** – Data dependence, unexpected events, overconfidence, ethical risks
- **Control and Robotics** – Limited adaptability, environmental challenges, safety risks, high costs

Ethics and Responsibility

- **Bias and Fairness:** AI can reproduce social inequalities if trained on biased data.
- **Privacy:** AI may misuse personal data without consent.
- **Transparency and Accountability:** “Black box” systems make it hard to explain or challenge decisions.
- **Dependence and Displacement:** Overuse of AI can replace jobs and reduce human skills.
- **Why Ethics Matters:** Ensures AI respects human rights, builds trust, and supports fairness, safety, and positive social change.

Activity 4.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	Narrow AI can only perform specific tasks and cannot transfer skills to other tasks.
<u>F</u>	General AI has already been achieved and is widely used in real-world applications.
<u>T</u>	Perception in AI refers to converting raw data like images and sounds into usable information.
<u>F</u>	Language AI is mainly about understanding human emotions through language.
<u>T</u>	Reasoning and Decision-Making AI evaluates options and selects actions, such as in recommender systems and logistics planning.
<u>T</u>	Prediction and Forecasting AI uses past data to estimate future outcomes, such as stock trends or weather forecasts.
<u>F</u>	Control and Robotics represents the input stage of AI, focusing on data collection from sensors.
<u>T</u>	One limitation of AI perception is fragility, meaning small changes in lighting or noise can cause errors.
<u>F</u>	AI systems are always transparent and easy to explain, which ensures accountability.
<u>T</u>	Ethical concerns include bias and fairness, privacy, transparency, and the risk of job displacement.

Activity 4.2

Answer the questions below.

1. Which of the following best describes Narrow AI?
 - a) AI that can learn any task like a human
 - b) **AI that can only perform specific tasks**
 - c) AI that always understands meaning
 - d) AI that replaces human judgment completely
2. What is General AI?
 - a) Current real-world AI used in translation tools
 - b) **A future AI that could learn and apply skills across many tasks**
 - c) An AI limited to perception and prediction only
 - d) An AI already used in self-driving cars
3. Which cognitive category involves converting raw data like pixels and sound waves into usable information?
 - a) **Perception**
 - b) Prediction and Forecasting
 - c) Reasoning and Decision-Making
 - d) Language
4. Which is an example of AI in Language processing?
 - a) Stock forecasting
 - b) Self-driving cars
 - c) **Translation tools**
 - d) Robotic surgery
5. AI used in recommender systems, logistics planning, and dynamic pricing belongs to which cognitive category?
 - a) Perception
 - b) **Reasoning and Decision-Making**
 - c) Control and Robotics
 - d) Language

Activity 4.2 (continued)

Answer the questions below.

6. What does Prediction and Forecasting AI mainly do?
 - a) **Uses past data to estimate future outcomes**
 - b) Summarizes large documents
 - c) Converts sound waves into text
 - d) Decides which move to play in a game

7. Which example belongs to Control and Robotics?
 - a) Detecting faces in photos
 - b) Chatbots for customer service
 - c) Weather forecasting
 - d) **Self-driving cars**

8. Which is a limitation of AI perception?
 - a) Errors in translation
 - b) **Fragility to small changes**
 - c) Hidden bias in hiring
 - d) Overconfidence in forecasts

9. Why is transparency and accountability an ethical concern?
 - a) AI always explains its decisions
 - b) AI systems are inexpensive and adaptable
 - c) AI never makes mistakes if data is good
 - d) **“Black box” systems make it hard to explain or challenge outcomes**

10. Which of the following is NOT listed as an ethical concern of AI?
 - a) Bias and fairness
 - b) Privacy
 - c) Job displacement
 - d) **Faster internet speeds**

Activity 4.3

Read the narrative and answer the questions below.

- a. A new security system at a train station uses AI cameras to detect suspicious items. One morning, a child leaves behind a toy water gun. The AI system quickly flags it as a potential weapon, causing the station to be evacuated. Later, officials realize it was only a harmless toy.

1. Which cognitive function of AI is being used in this scenario?

Perception – analyzing visual input from cameras.

2. Why did the system mistake the toy water gun for a real weapon?

It recognized the shape but not the context.

3. What limitation of AI perception does this scenario highlight?

AI perception lacks common sense.

4. How could human oversight have prevented the unnecessary evacuation?

A human officer could have reviewed the flagged item and recognized it was only a toy before ordering an evacuation.

- b. An international company uses an AI translation tool for customer support. One customer in America writes, “It’s raining cats and dogs here, will that make my delivery late?” The AI translates the phrase literally into Mandarin. The customer support person reads it and is confused: “How could it rain cats and dogs?!”

1. Which cognitive function of AI is being used in this scenario?

Language – processing and translating text.

2. Why did the translation confuse the customer support person?

The AI translated the idiom word-for-word.

3. What limitation of AI language does this scenario highlight?

AI language has trouble with context and idiomatic expressions.

4. How could the company improve the accuracy of its translations?

By using human review or improving the system with more cultural/linguistic data.

Activity 4.3 (continued)

Read the narrative and answer the questions below.

- c. A food delivery company uses AI to assign drivers to orders. Drivers notice that short, high-paying deliveries mostly go to certain neighborhoods, while other areas get fewer and less profitable orders. The company later admits the AI was trained on past data that favored wealthier neighborhoods.

1. Which cognitive function of AI is being used in this scenario?

Reasoning and Decision-Making – assigning deliveries.

2. What unfair pattern did the drivers notice in the AI's decisions?

Wealthier neighborhoods got more profitable orders.

3. What limitation of AI reasoning does this scenario highlight?

Hidden bias in the training data.

4. How might the company fix this problem?

Retrain with fairer data and set rules for equal distribution.

- d. A city uses AI to forecast traffic for road repairs. The system predicts light traffic for a weekend, even though a major sports event is scheduled. Because the AI had never seen such an event before, it ignored it. Human planners assumed the AI accounted for everything, so construction went ahead. On game day, thousands of fans arrived, and the city was gridlocked.

1. Which cognitive function of AI is being used in this scenario?

Prediction and Forecasting – estimating future traffic.

2. Why did the prediction fail, even though the AI had data about the sports event?

It had no prior examples of such an event, so it ignored the data.

3. What limitation of AI prediction does this highlight?

Inability to handle the unexpected.

4. How could humans have helped avoid the problem?

By being aware of AI's limits, the planners should have reviewed the forecast critically and applied human judgment for unusual events.

Activity 4.3 (continued)

Read the narrative and answer the questions below.

- e. A city tests self-driving buses on public roads. Most of the time, the buses run smoothly. One day, however, a pedestrian suddenly crosses the street outside a crosswalk. The bus hesitates for a moment before stopping, narrowly avoiding an accident. City officials realize the AI system was not fully prepared for unpredictable human behavior.

1. Which cognitive function of AI is being used in this scenario?

Control and Robotics – managing self-driving buses.

2. What nearly caused the accident?

The AI hesitated when faced with unexpected pedestrian behavior.

3. What limitation of AI control and robotics does this highlight?

Safety risks – AI-controlled machines may fail in unpredictable real-world situations.

4. How could humans help reduce this type of risk?

By being aware of AI's limits and ensuring human oversight or emergency controls are in place.

- f. A tech company announces a new AI that can drive a car, write novels, prepare food, and give medical advice—all without retraining. Many people line up to buy this amazing new technology. Would you buy it, or would you be skeptical?

1. What type of AI is being claimed in this scenario?

General AI – an AI that can perform many different tasks across domains.

2. Why might people be excited about such a system?

Because it promises to combine many human-like abilities into one system.

3. Why should we be skeptical of this claim today?

We should be skeptical because General AI has not yet been achieved; all existing AI is still Narrow AI.



Activity 4.4 - Exploring AI in the Real World

In this activity, you will interact with an AI which will act as a mentor, providing examples, questions, and explanations to deepen your understanding of AI applications, limitations, and ethics. In the shared folder you will find separate text documents, each containing the prompt you need for the activity, as well as the PDF of the lesson summary.

Access the folder from: bit.ly/C4_AI_Activity4_4

a. Scenario Exploration

In this part, you will practice identifying AI cognitive functions and their limitations through real-world scenarios.

- Perception
- Language
- Reasoning & Decision-Making
- Prediction & Forecasting
- Control & Robotics

Steps to Follow:

1. Open the shared folder and download the PDF of the lesson summary.
2. Attach the PDF in ChatGPT (or another LLM).
3. Open the Scenario Exploration file, then copy the full prompt and paste it into ChatGPT.
4. You will receive one short story scenario at a time, each showing a real-world application of AI in one of the five cognitive functions.
5. Each scenario hides a limitation or challenge for you to uncover.
6. After the scenario, you will be asked four comprehension questions, one at a time. Answer each question, and the AI will tell you if you are correct, explain briefly, and correct you if necessary.
7. At the end of each scenario, the AI will review the questions you got wrong, highlight key topics you struggled with, and suggest one follow-up exercise.
8. The next scenario will move on to a different cognitive function (no repeats). Continue until all five functions are covered.

Always explain your reasoning, even if unsure — that is what the AI will evaluate.

Don't rush.

b. Questions & Answers

In this part, you will test your understanding of the lesson using the PDF provided in the folder. You can choose between:

- True/False Questions
- Multiple-Choice Questions

Each activity has its own text file with a prompt.

Steps to Follow:

1. Open the shared folder and download the PDF of the lesson summary.
2. Attach the PDF in ChatGPT.
3. Copy the prompt for the activity you want to practice (True/False or Multiple Choice) and paste it into ChatGPT.
4. The AI will create 10 questions based on the summary:
 - If you chose True/False, each question will be answered with “True” or “False.”.
 - If you chose Multiple Choice, each question will have 4 options (A, B, C, D).
5. For each question:
 - Read carefully and submit your answer.
 - The AI will tell you if you are correct, provide the correct answer if needed, and give a short explanation.
6. After all questions:
 - The AI will list the questions you got wrong.
 - It will give you a short review of the key topics you struggled with.
 - It will suggest a follow-up exercise to strengthen your understanding.

Tips:

- For True/False: think about why the statement is true or false before answering.
- For Multiple Choice: read all four options carefully before choosing.
- Use the AI’s feedback to focus on the concepts you find most difficult.
- Repeat the activity for more practice — the AI can generate new sets of questions each time.

c. Ethics Debate

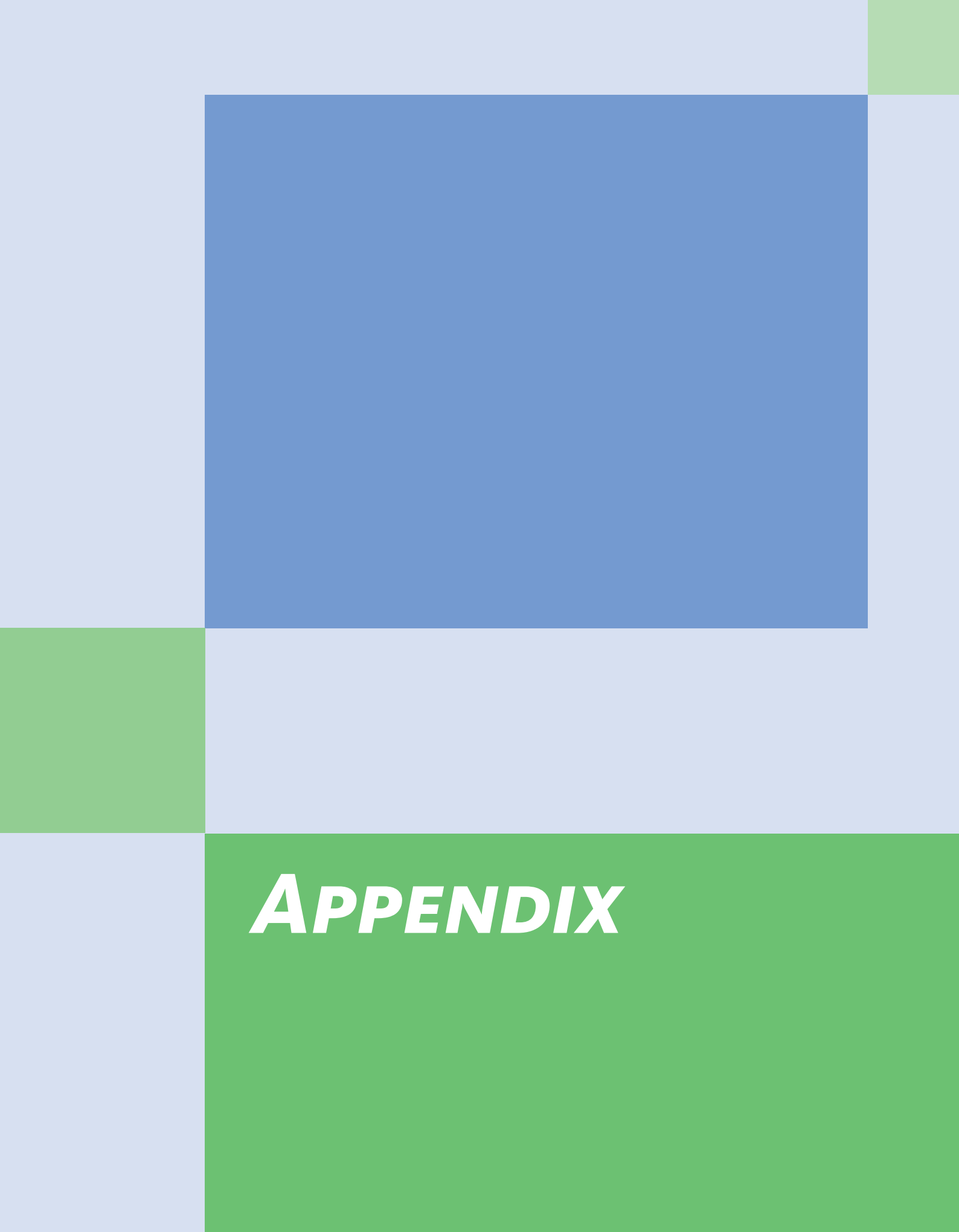
This optional part lets you practice thinking critically about AI and society.

Steps to Follow:

1. Copy the Ethics Debate Prompt into ChatGPT.
2. The AI will first list the available debate themes for you to choose from: Bias, Privacy, Transparency, Job Displacement.
3. After you select a theme, the AI will give you one ethical scenario for it.
4. You will then be asked 3–4 debate questions, one at a time. After each answer, the AI will tell you if your response is strong or weak, challenge you with a counterargument, and suggest improvements.
5. At the end of the debate, the AI will give you a short review of the ethical concepts and suggest one follow-up exercise.
6. Finally, the AI will ask if you want to try another theme. You may continue until you've practiced as many themes as you like.

Tips:

- There's no single "right" answer. Focus on reasoning clearly.
- Use evidence from the lesson (bias, fairness, human oversight).

The background of the page is a light blue grid. A large blue rectangle is positioned in the upper left quadrant. A green rectangle is in the upper right corner. A green rectangle is on the left side, below the blue rectangle. A large green rectangle occupies the bottom right portion of the page, containing the word 'APPENDIX' in white.

APPENDIX

GLOSSARY

TERM	DEFINITION	PAGE
Accuracy	A measure of how often a model's predictions are correct, calculated as the percentage of correct predictions out of the total.	p. 177
Actionable Information	Information that has been interpreted and can be directly used to make informed decisions.	p. 21
Adjacency List	A way to represent a graph by listing each node and the nodes it connects to, along with edge information.	p. 32
Algorithm	A step-by-step set of instructions that tells a computer how to solve a problem or perform a task. Algorithms can manage data, search through it, or arrange it in order.	p. 33, 68, 139
Array	A linear data structure where items are stored one after another in a single line. Arrays have a fixed size and usually hold items of the same type, so new items cannot be added or removed once created.	p. 12, 33
Artificial General Intelligence (AGI / General AI)	A theoretical form of AI that could learn and apply intelligence across many different tasks, like a human.	p. 217
Artificial Intelligence (AI)	The field of computer science focused on building machines that can perform tasks that normally require human intelligence.	p. 159
Ascending Order	Sorting data from smallest to largest or from A to Z.	p. 95
Association	A link formed between two ideas, such as connecting the image of a cup with the word "cup."	p. 158
Bar Chart	A chart that represents values using vertical or horizontal bars, useful for comparing quantities across categories.	p. 15
Best-Case Scenario	The situation where an algorithm finishes its task in the fewest possible steps (e.g., finding the item on the first try).	p. 73
Bias	An unfair pattern in AI predictions or decisions caused by biased training data or flawed design.	p. 228
Big Data	Large and varied sets of data that help AI models learn better and make more reliable predictions.	p. 184
Binary Search	A search algorithm for sorted datasets that repeatedly divides the list in half to find the target more quickly.	p. 75
Black Box	A system, often in AI, where the decision-making process is hidden and hard to explain or challenge.	p. 223
Block (in programming)	A section of code grouped together to perform a task; examples include initialization, main body, and termination.	p. 140
Bubble Sort	A sorting algorithm that repeatedly compares and swaps adjacent items until the list is sorted.	p. 111

Categorical Data	Another term for qualitative data; data that can be grouped and labeled but not measured.	p. 18
Categorical Prediction	A type of prediction where the AI assigns inputs to categories, such as identifying if an object is a bottle or not.	p. 175
Chart	A visual method of presenting data so patterns and trends can be easily seen (e.g., line, bar, pie, scatter).	p. 14
Child Node	A node in a tree that descends from another node (its parent).	p. 46
Cleaning Data	The process of removing errors or incorrect entries from a dataset so the data is accurate.	p. 12
Compare, Shift, Insert	The basic strategy used by insertion sort: compare items, shift larger ones, and insert the held item in the right place.	p. 99
Continuous Data	Quantitative data that can take any value within a range, such as temperature or time.	p. 16
Continuous Learning	The idea that a model keeps improving over time as it encounters new data and situations.	p. 165
Continuous Prediction	A type of prediction where the AI estimates a numerical value, such as house prices or temperatures.	p. 178
Control & Robotics	The use of AI to direct machines in the physical world, such as self-driving cars or delivery robots.	p. 226
Criterion (Sorting Criterion)	The rule or property used to decide how to sort data, such as by score, name, or price.	p. 95
Data	Raw input that has not yet been processed to add meaning.	p. 8
Data Structure	The specific way in which data is arranged and stored in a computer.	p. 12
Decision-Making	The process of selecting the best choice from several options by comparing their outcomes and choosing the one with the highest utility.	p. 197
Dependence & Displacement	When overreliance on AI reduces human roles or causes job loss without proper support or retraining.	p. 228
Descending Order	Sorting data from largest to smallest or from Z to A.	p. 95
Discrete Data	Quantitative data that is countable and can only take whole number values, such as number of students.	p. 17
Dynamic Size	A feature of some data structures (like lists, stacks, and queues) that allows them to grow or shrink as items are added or removed.	p. 34
Edge	In a graph, a link between two nodes that can carry extra information, such as distance or time.	p. 50
Error	The difference between the predicted value and the actual value.	p. 180
Ethics (in AI)	Principles that guide AI to be fair, safe, and beneficial to society while avoiding harm.	p. 228

Execution Time	The amount of time it takes for a program or algorithm to complete its run. It is often used to compare how efficiently different algorithms perform.	p. 72, 124
Explanation	A form of reasoning that works backward from an outcome (a given) to find the most plausible cause.	p. 201
FIFO (First In, First Out)	A principle where the first item added is the first one removed, used in queues.	p. 42
Fixed Size	A feature of some data structures (like arrays and tuples) that means their size cannot change once created.	p. 33
Flow (of a program)	The sequence in which steps of an algorithm or program are carried out.	p. 139
Flowchart	A diagram that shows the steps of a process using symbols and arrows to indicate decisions and actions.	p. 69
General AI	See Artificial General Intelligence (AGI).	p. 217
Generalization	The ability of a model to apply what it has learned to correctly identify or handle new, unseen examples.	p. 163, 180
Given	The starting point of explanation, an accepted observation or outcome that needs to be explained.	p. 201
Graph	A non-linear data structure where nodes are connected by edges, allowing many-to-many relationships.	p. 32
Hierarchical Structure	An arrangement where items are organized in levels, like in a tree with parent and child nodes.	p. 31
Indentation	The spacing used in code to show which instructions belong to a block or loop.	p. 140
Index	The position number of an item in a list or array, usually starting at 0. It helps locate and access items in a linear data structure.	p. 35, 70
Inference	A form of reasoning that links known facts, observations, or predictions to reach a new conclusion.	p. 195
Information	Data that has been processed and given meaning, making it useful for understanding or decision-making.	p. 8
Inner Loop	The loop inside another loop; in sorting, it compares and shifts or swaps data items repeatedly during each pass.	p. 104, 117
Insertion Sort	A sorting algorithm that builds a sorted section one item at a time by comparing, shifting, and inserting.	p. 99
Interpreting Data	The step of explaining what data means, asking “why” it happened and “what should we do.”	p. 20–21
Knowledge Base	The collection of rules, facts, or learned patterns that reasoning relies on to connect conditions to outcomes.	p. 206
Labeling Data	Adding descriptive tags or identifiers (like “temperature in °C”) to clarify what the data represents.	p. 11

Language (AI)	AI systems that process, understand, or generate human language in text or speech.	p. 220
Leaf Node	A node in a tree that has no children.	p. 46
Learning	The ability to form new associations and make use of them through repetition and pattern recognition.	p. 158
LIFO (Last In, First Out)	A principle where the last item added is the first one removed, used in stacks.	p. 39
Line Chart	A chart where data points are connected by a line, often used to show trends over time.	p. 14
Linear Data Structure	A type of data structure where items are arranged sequentially, like arrays, lists, stacks, and queues.	p. 30
Linear Search	A simple search algorithm that checks each item in a list one by one until it finds the target or reaches the end.	p. 68
List	A linear data structure with a dynamic size that can store mixed data types and allows insertion and removal at any position.	p. 36
Logarithmic Growth	A type of growth where progress increases quickly at first but slows as the dataset gets larger (common in binary search).	p. 80–81
Logic	The rules that guide reasoning and ensure conclusions follow consistently from premises.	p. 194
Loop	A control structure that repeats a set of instructions until a condition is met.	p. 104
Machine Learning	A branch of AI where a computer learns from data by recognizing patterns and forming associations.	p. 159
Mean Absolute Error (MAE)	A way to measure how accurate a model's numerical predictions are, by calculating the average size of the errors (ignoring whether they are positive or negative).	p. 180–181
Memory Latency	The delay between requesting data from memory and receiving it.	p. 140
Misclassification	A prediction error where a model gives the wrong output, such as failing to recognize a cup with a handle or labeling a tomato as an apple.	p. 163, 177
Model	The AI's internal system that stores recognized patterns and associations, used to make predictions or classifications.	p. 162
Narrow AI	AI designed for a specific task or limited set of tasks, such as translation or facial recognition.	p. 216
Natural Language Processing (NLP)	A branch of AI that analyzes, interprets, and generates human language.	p. 220
Node	A unit of data in tree or graph structures that can connect to other nodes.	p. 46
Non-Linear Data Structure	A type of data structure where data items are connected in more complex ways, like trees and graphs.	p. 31

Organizing Data	Arranging data in ways such as cleaning, structuring, or sorting so it becomes easier to analyze.	p. 11
Outer Loop	The main loop that controls how many passes through the list an algorithm makes.	p. 105, 118
Pair, Compare, Swap	The basic strategy used by bubble sort: compare pairs of adjacent items and swap them if needed.	p. 111
Parent Node	A node in a tree that has one or more child nodes.	p. 46
Pattern Recognition	The ability of a system to detect recurring regularities, such as shapes or sounds that repeat.	p. 160
Perception (AI)	The ability of AI to interpret sensory input, like vision or sound, using sensors, cameras, or microphones.	p. 217
Performance	A measure of how efficiently an algorithm runs, often judged by the number of steps it takes and the time it needs to finish.	p. 72, 124
Pie Chart	A circular chart divided into slices that represent proportions of a whole.	p. 15
Plausibility	A measure of how believable or reasonable an explanation is, based on how well its premises fit the outcome.	p. 201
Prediction	When an AI system uses its model to estimate the most probable outcome in a new situation.	p. 174
Prediction & Forecasting (AI)	Using past and current data to identify patterns and estimate future outcomes.	p. 224
Premise	The starting point of reasoning, which can be a fact, observation, or prediction.	p. 195
Privacy	The right to keep personal data protected from misuse or unauthorized access.	p. 228
Qualitative Data	Descriptive data that can be categorized into groups (like colors or gender) but not measured.	p. 18
Quantitative Data	Numerical data that can be measured or counted.	p. 16
Queue	A linear data structure with a dynamic size where items are added at the end and removed from the beginning, following FIFO.	p. 42
Reasoning	A structured way of thinking that helps us discover, decide, or explain by following logical rules.	p. 194
Reasoning & Decision-Making (AI)	AI systems that evaluate options and choose actions within a defined area, like recommender systems.	p. 222
Root Node	The top node in a tree, from which all other nodes descend.	p. 31, 46
Scalability	The ability of an algorithm or AI system to keep performing well as the dataset grows larger or the tasks become more complex.	p. 130, 225
Scatter Plot	A chart that shows the relationship between two continuous variables as individual points.	p. 14
Search Index	The current position in a dataset that an algorithm is checking.	p. 78

Search Range	The section of a dataset that an algorithm is currently looking at during a search.	p. 78
Sentiment Analysis	An AI tool that scans reviews, posts, or text to detect opinions, such as positive, negative, or neutral.	p. 220
Sorted List	A list in which items are arranged in order (such as numbers from smallest to largest or words alphabetically).	p. 75
Sorting	The process of arranging data in a specific order based on a chosen criterion.	p. 94
Sorting Data	Re-arranging data according to a criterion such as alphabetical order, numerical order, or date.	p. 13
Speech Recognition	An AI capability that converts spoken words into text.	p. 218
Stack	A linear data structure with a dynamic size where items can only be added or removed from the end, following LIFO.	p. 39
Step Count	The number of individual operations an algorithm performs while completing its task.	p. 124
Structuring Data	Storing data in an organized way, such as in lists (arrays) or tables.	p. 12
Supervised Learning	A type of machine learning where the system is trained with labeled data (e.g., images tagged as “cup”).	p. 160
Swap	An operation where two data items exchange positions, usually requiring three moves in memory.	p. 135
Table	A way of structuring data using rows and columns to handle multiple sets of related data.	p. 12–13
Target	The specific item an algorithm is trying to find in a dataset.	p. 70
Termination	The end block of a program or algorithm that signals completion of the task.	p. 110, 123
Training	The process of improving a model using new data and correcting mistakes.	p. 163
Training Data	The examples used to teach an AI model, helping it learn patterns for making predictions.	p. 178
Transparency & Accountability	The principle that AI systems should explain their decisions clearly so they can be reviewed and corrected if needed.	p. 228
Tree	A non-linear data structure that organizes items hierarchically in nodes connected as parents and children.	p. 31
Tuple	A linear data structure with a fixed size and format that can hold different types of data, often pairing related data together.	p. 37
Unsupervised Learning	A type of machine learning where the system groups data based on similarities without using labels.	p. 161
Utility	A measure of how useful, valuable, or goal-fitting an outcome is, used in decision-making.	p. 200
Variable	A label in programming that holds data and whose value can change.	p. 103

Variety (in data)	Having many different types of examples in the training data, which helps the model make better predictions in new situations.	p. 184
Visualizing Data	Presenting data in a visual format like charts or graphs to make trends and patterns clearer.	p. 14
Volume (in data)	The total amount of training data available to the model, which helps it learn broader patterns.	p. 184
Weight	Extra information stored on a graph edge, such as the distance or time between two nodes.	p. 51
Worst-Case Scenario	The situation where an algorithm takes the maximum possible steps to complete a task (e.g., checking the whole list).	p. 73

INDEX

Accuracy – Measure of performance for categorical predictions; percentage of correct predictions out of total attempts (p.176–178).

Adjacency list – A way of representing a graph by listing each node along with the nodes it connects to and the weight of each edge (p.32, 50–52).

Algorithm – A series of steps designed to perform a specific task, such as sorting a list (p.139–140).

- Algorithm performance – Evaluation of how efficiently an algorithm runs, often measured by execution time in best-case and worst-case scenarios (p.72–74).
- Algorithms (data structures) – Sets of instructions that control data structures, defining rules for size, data type, and operations (p.33).

Artificial General Intelligence (AGI / General AI)

– A theoretical AI system that could learn and apply intelligence flexibly across many tasks, similar to humans. Not yet achieved (p.217, 236).

Associations – Links formed between known and new information; essential for learning (p.158–159).

Array (List) – A linear data structure that stores items sequentially in memory. Arrays have a fixed size and homogeneous data type: items can be updated but not inserted or removed (p.12–13, 33–35).

Bar chart – A chart in which categories are represented by bars; useful for comparing quantities across categories. Can be vertical or horizontal (p.15, 17, 19).

Bias – An ethical concern where AI reproduces unfair patterns from biased training data, leading to unequal treatment (p.228–229).

Best-case scenario – The situation in which an algorithm finds the target immediately or completes its task with the fewest possible steps. For example, in search, it may succeed on the very first item checked (p.73, 85).

Big Data – Large datasets characterized by volume and variety; improve AI models by helping them generalize

and make more accurate predictions (p.184).

Binary search – Search algorithm for sorted datasets that repeatedly divides the search range in half until the target is found or eliminated (p.75–79).

- Flowchart – Diagram showing binary search process (p.77).
- Pseudocode – Step-by-step outline of binary search (p.79).
- Execution time – Grows logarithmically with dataset size, modeled with $\log_2(n)$ (p.80–82).
- Requirements – Dataset must be sorted (p.75, 86).

Black box problem – A limitation of AI reasoning and decision-making where systems lack transparency, making it difficult to explain or challenge outcomes (p.223).

Block – A segment of code that performs a task, usually part of initialization, main body, or termination (p.140).

Bubble sort – Sorting algorithm using the strategy of pair, compare, and swap. Repeatedly passes through the list, swapping adjacent items if out of order (p.111–113, 117–123).

- Process – Uses outer loop (repeats passes) and inner loop (compares/swaps adjacent pairs) (p.117–123).
- Pseudocode – Instructions for bubble sort (p.120–123, 153).
- Performance – Requires more steps than insertion sort; poor scalability; inefficient for large datasets (p.134–138).

Categorical data – Another name for qualitative data, which can be grouped and labeled but not measured (p.18).

Categorical prediction – Prediction task where the AI classifies inputs into categories (e.g., cat or not a cat, language detection) (p.175–176).

Charts – Visual tools to present data. Four common types: line chart, scatter plot, bar chart, and pie chart

(p.14–15).

Child node – A node that descends from a parent node in a tree structure (p.46).

Cleaning data – Removing erroneous or implausible data values before processing (p.12).

Continuous data – Quantitative data that can take any value within a range (e.g., temperature, height, time). Best visualized with line charts or scatter plots (p.16–17).

Continuous learning – The idea that learning does not stop after training; models can keep improving with new data and feedback (p.165).

Continuous prediction – Prediction task where the AI estimates a numerical value (e.g., house prices, rainfall, travel times) (p.178–180).

Control and Robotics – The use of AI to direct machines acting in the physical world. Applications include self-driving cars, industrial robots, delivery drones, robotic surgery, agricultural robots, and disaster-response robots (p.226–227).

Criterion (sorting) – The basis for sorting, such as numeric (scores, prices, time) or alphanumeric (names, book titles) (p.95).

Data – Raw values or signals that can be processed into meaningful information (p.8).

Data structures – Methods of organizing data in computers, such as lists (arrays), tables, stacks, queues, trees, and graphs (p.12–13).

- Dynamic size – Property of structures like lists, stacks, and queues that allows them to grow or shrink as items are added or removed (p.34, 39, 42).

Decision-making – The process of selecting the best choice from several options by weighing their outcomes and choosing the one with the highest utility (p.197–200).

Discrete data – Quantitative data that can only take whole-number values (e.g., number of students). Often visualized using bar charts (p.17).

Edge – A link between two nodes in a graph, often carrying additional information such as distance or latency (p.32, 50–52).

Error – The difference between predicted and actual values; can be positive (underestimation) or negative (overestimation) (p.180).

Ethics and responsibility – Principles ensuring AI respects human rights, fairness, safety, privacy, and accountability, while reducing risks of harm or inequality (p.228).

Execution time – The time required for a program to complete its run.

- In search – Calculated as time per attempt $\times$ number of attempts (p.72, 80–82).
- In sorting – Influenced by step count and memory latency (p.124–125, 131–133, 138).

Explanation – Reasoning backward from an observed outcome to uncover its most plausible causes, contributors, or relationships (p.201–204).

FIFO (First In, First Out) – Principle governing queues; the first item added is the first removed (p.42–44).

Flow & control – Flow is the sequence of steps in a program; control refers to conditions that determine which steps are executed, repeated, or skipped (p.139).

General AI – see Artificial General Intelligence (AGI).

Given – An accepted observation or outcome that serves as the starting point for explanation (p.201–202).

Graph – A non-linear data structure where nodes can connect freely to represent complex relationships. Edges may store additional information (p.32, 49–54).

Hierarchy – Organizational model in which data items are arranged at levels, such as in a tree structure (p.31, 45–47).

Hold index / hold data – Variables in sorting algorithms representing the index and the value temporarily stored for comparison (p.103, 116).

Information – Data that has been processed into meaningful form, typically through identification, organization, visualization, and interpretation (p.8–11, 21).

- Interpreting data – The final step; answering “why” and “what should we do?” to make data actionable (p.20–21).

- Labeling data – Adding descriptive identifiers (e.g., “temperature, outdoor, Celsius”) to clarify meaning and context (p.11).
- Organizing data – Arranging data to make it usable, through cleaning, structuring, or sorting (p.11–13).

Inference – Arriving at a new piece of information by linking premises (facts, observations, or predictions) into a conclusion (p.195–197).

Initialization – The first block of a program, where variables are declared and initial values assigned (p.106–107, 119–120).

Insertion sort – Sorting algorithm using the strategy of compare, shift, and insert. Builds a sorted section one item at a time (p.99–106, 139–140).

- Process – Uses outer loop (moves through each index) and inner loop (compares/shifts until correct position is found) (p.103–106).
- Pseudocode – Instructions for insertion sort (p.106–110, 152).
- Performance – Fewer steps and faster than bubble sort; still scales poorly for very large datasets (p.124–130).

Job displacement – Risk of AI replacing human workers without retraining opportunities, leading to unemployment and inequality (p.229).

Knowledge base – The collection of rules, facts, and learned patterns that reasoning processes rely on to connect conditions to outcomes (p.201).

Labeling data – see Information.

Language (AI cognitive function) – Natural Language Processing (NLP) enables AI to analyze, interpret, and generate human language. Applications include translation tools, chatbots, large language models, sentiment analysis, and educational tools (p.220–221).

Leaf node – A node in a tree with no child nodes (p.46).

Learning – The ability to form and use new associations; requires pattern recognition, association, and repetition (p.158–159, 166).

LIFO (Last In, First Out) – Principle governing stacks; the last item added is the first removed (p.39–41).

Linear search – A simple search algorithm that checks each item in order until the target is found or the end is reached (p.68–71).

- Flowchart – Diagram showing linear search process (p.69–70).
- Pseudocode – Step-by-step outline of linear search (p.71).
- Execution time – Grows linearly with dataset size (p.72–73).
- Applicability – Works on both sorted and unsorted datasets (p.75, 84).

Linear structures – Data structures where items are stored sequentially: array, list, tuple, stack, queue (p.33–34).

List – A linear data structure with dynamic size, accepts mixed data types, and allows insertion or removal at any position (p.36–37).

Logic – The rules that guide reasoning from premises to conclusions; the framework for inference, decision-making, and explanation (p.194).

Logarithmic growth – Growth pattern of binary search execution time; efficiency improves as dataset size increases (p.80–82).

Loop – Control structure used to repeat a set of instructions. Sorting algorithms use outer and inner loops (p.104–106, 117–122).

Machine learning – A branch of AI where computers are designed with learning capabilities to identify patterns, form associations, and improve through repetition (p.159).

Mean Absolute Error (MAE) – A measure of performance for continuous predictions; the average size of errors regardless of direction (p.180–183).

Memory latency – Delay between requesting data from memory and receiving it; becomes significant for large datasets stored outside cache (p.131–133, 138–139).

Misclassification – An error where a model incorrectly labels or classifies input data (p.163–164, 176–177).

Model – The internal system that represents recognized patterns, labels, and rules about what features matter or don’t (p.162–163).

Natural Language Processing (NLP) – AI technology that processes, translates, and generates human language in text or speech (p.220).

Narrow AI – Current form of AI designed to perform a specific task or a small set of tasks very effectively (p.216–217).

Node – A data construct used in trees and graphs to store both data and links to other nodes (p.46–48, 50).

Order (sorting) – Direction in which data is arranged, e.g., ascending (least to greatest) or descending (greatest to least) (p.95).

Parent node – A node with child nodes branching from it in a tree structure (p.46).

Pattern recognition – The ability to detect recurring regularities, such as the shared shape of cups regardless of color or design (p.160).

Pie chart – A circular chart showing categories as slices of a whole, indicating proportions or percentages (p.15, 20).

Plausibility – A measure of how reasonable or believable an explanation is, based on how well its premises fit the observed outcome (p.201, 204).

Prediction – Process by which AI (or humans) estimates the most probable outcome based on learned patterns (p.174–175).

Prediction & Forecasting – AI's ability to analyze past data to estimate future outcomes. Applications include finance, healthcare, meteorology, retail, education, sports, and urban planning (p.224–225).

Premises – The starting points of reasoning; can be facts, observations, predictions, preferences, or rules (p.195, 197–198).

Privacy – An ethical issue involving AI misuse of personal or sensitive data without consent (p.229).

Program – A set of written instructions a computer can execute to run an algorithm (p.139).

Qualitative data – Descriptive, categorical data that can be observed but not measured (e.g., gender, grade levels, types of cars). Must often be converted into quantitative form for visualization (p.18–19).

Quantitative data – Data that can be measured (continuous) or counted (discrete). Best visualized with line charts, scatter plots, or bar charts (p.16–17).

Queue – A linear data structure that follows FIFO. Items are added at the end and removed from the beginning (p.42–44).

Reasoning – A structured way of thinking to discover, decide, or explain. Outcomes may be inference, decision, or explanation (p.194, 206).

Reasoning and Decision-Making (AI) – AI's ability to evaluate options and select actions in a specific domain. Examples include recommender systems, logistics planning, dynamic pricing, and game-playing AIs (p.222–223).

Repetition – Strengthens associations and improves pattern recognition in both humans and AI (p.158–159, 166).

Root node – The topmost node in a tree structure from which all other nodes descend (p.46).

Scalability – How an algorithm's performance changes as dataset size increases. Sorting algorithms like insertion and bubble sort scale poorly due to quadratic growth (p.129–130, 137–138).

Scatter plot – A chart showing individual data points, often used to highlight relationships or correlations between two variables (p.14, 16–17).

Search index – Position within a dataset being checked during a search algorithm (p.71, 78).

Search range – Portion of a dataset still under consideration in a binary search; updated each iteration (p.78).

Searching – Fundamental operation required to locate a specific item in a dataset before modification, extraction, or processing (p.68, 83).

Sorting – The process of arranging data in a specific order based on a criterion, such as chronological order, increasing or decreasing value, or categories. Sorting makes it easier to search, interpret, and analyze data (p.13–14, 94–98).

Stack – A linear data structure that follows LIFO. Items can only be added or removed from the end (p.39–41).

Step count – The number of operations an algorithm performs. Used alongside execution time to measure performance (p.124, 126–127, 135).

Supervised learning – A type of machine learning that uses labeled data; the system is trained with examples

where the correct answer is provided (p.160).

Swap – Operation in bubble sort where two adjacent items exchange positions. Each swap requires three moves in memory (p.135).

Table – An arrangement of data in rows and columns; well-suited for organizing multiple sets of related data (p.12–13).

Termination – The final block of a program where a message indicates the task is complete and the program ends (p.110, 123).

Training – The process of improving a model using new data and corrections. It involves analyzing mistakes, refining rules, and updating the model to improve accuracy and generalization. Effective training depends on both the volume (amount) and variety (diversity) of training data (p.163–165, 178, 183–184).

Transparency and accountability – Ethical principle requiring AI decisions to be explainable and challengeable. Lack of transparency creates risks of unfair or harmful outcomes (p.229).

Tree – A non-linear hierarchical data structure with nodes connected along branches, including root, parent, and child nodes (p.31, 45–48).

Tuple – A linear data structure with fixed size and format, can store multiple data types together as a unit (p.37–38).

Uncertainty – The degree to which reasoning outcomes may not be definite, especially when based on predictions or incomplete premises (p.197, 204–205).

Underestimation/Overestimation – When predictions consistently fall below or above actual values (p.180–182).

Unsupervised learning – A type of machine learning that uses unlabeled data; the system identifies patterns and groups similar items without predefined labels (p.161).

Utility – A measure of how useful, valuable, or goal-fitting an outcome is; used to evaluate decisions (p.200).

Visualization – Presenting data in visual form (charts, graphs) to highlight patterns and trends (p.14–15).

Worst-case scenario – Situation where an algorithm must check all possible items before concluding, or in sorting, when data requires the maximum steps (e.g., reversed order for ascending sort). Represents the longest possible execution time (p.73–74, 124–127).

